

Cloud Based Data Engineering

Approximation and Randomization

Expectation

Let X be a discrete random variable with values $x_1, x_2, \dots, x_n$.

$$E[X] = \sum_{i=1}^n x_i \cdot P(X = x_i)$$

Linearity: ALWAYS, no matter what are the relations between X and Y !

$$E[X + Y] = E[X] + E[Y]$$

$$E[cX] = cE[X]$$

Variance

Let X be a discrete random variable with values $x_1, x_2, \dots, x_n$.

Variance:

$$V[X] = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

$$V[cX] = c^2 V[X]$$

Independence

Let X, Y be random variables, then X, Y are independent if and only if

$$P[X = x_i, Y = y_i] = P[X = x_i] \cdot P[Y = y_i]$$

Probability Tools

Union Bounds: Let $E_1, E_2, \dots, E_n$ events, then

$$P[E_1 \cup E_2 \cup \dots \cup E_n] \leq P[E_1] + P[E_2] + \dots + P[E_n]$$

Bernoulli's inequality:

$$1 + x \leq e^x$$

Sum of arithmetic series:

$$S_n = \frac{n}{2} [a_1 + a_n]$$

Collision Probability

Using hash-functions in streaming algorithms, require analysis of collisions.

For better story, we will use the problem:

We have m balls and n bins.

Each step, one balls is throw to a bin.

Since we want to **bound** the probability, we may use Bernoulli's inequality:

$$1 + x \leq e^x$$

Collision Probability

We have m balls and n bins. Each step, one balls is throw to a bin.

If $n > m$, find the collision probability (a ball is land in a bin that already has at least 1 ball).

Events: $i = 1, 2, \dots, m: A_i$ - no collision when the i^{th} ball is thrown

$$P[A_1] = 1$$

$$P[A_2|A_1] = \frac{n-1}{n} = 1 - \frac{1}{n}$$

$$P[A_3|A_2, A_1] = \frac{n-2}{n} = 1 - \frac{2}{n}$$

Generally:

$$P[A_i | A_{i-1}, \dots, A_1] = 1 - \frac{i-1}{n}$$

$$P[\text{no-collision}] = A_1 \cap A_2 \cap \dots \cap A_m$$

All events are independent:

$$\begin{aligned} P[A_1 \cap A_2 \cap \dots \cap A_m] &= P[A_1] \cdot P[A_2 | A_1] \cdots P[A_m | \cap_{i=1}^{m-1} A_i] = \\ &= \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{m-1}{n}\right) = \prod_{i=1}^{m-1} \left(1 - \frac{i}{n}\right) \end{aligned}$$

Using $1 + x \leq e^x$ (Bernoulli's inequality):

$$\left(1 - \frac{1}{n}\right) \leq e^{-\frac{1}{n}}$$

$$\left(1 - \frac{2}{n}\right) \leq e^{-\frac{2}{n}}$$

...

$$\left(1 - \frac{m-1}{n}\right) \leq e^{-\frac{m-1}{n}}$$

We get:

$$P[A_1 \cap A_2 \cap \dots \cap A_m] = \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{m-1}{n}\right) \leq$$

$$e^{-\frac{1}{n}} \cdot e^{-\frac{2}{n}} \dots e^{-\frac{m-1}{n}} = e^{-\frac{1}{n} - \frac{2}{n} - \dots - \frac{m-1}{n}}$$

$$-\frac{1}{n} - \frac{2}{n} - \dots - \frac{m-1}{n} = -\frac{1}{n}(1 + 2 + \dots + (m-1))$$

Arithmetic series $S_n = \frac{n}{2} [a_1 + a_n]$

$$-\frac{1}{n} \left[\frac{m-1}{2} (1 + m-1) \right] = -\frac{m(m-1)}{2n}$$

$$P[\text{no collision}] \leq e^{-\frac{m(m-1)}{2n}}$$

Now we can decide what number of bins is the best to minimize collisions 😊

Approximation and Randomization

Many things are hard to compute exactly over a stream

- Is the count of all items the same in two different streams?

Approximation: find an answer correct within some factor

“Find an answer that is within 10% of correct result”

Randomization: allow a small probability of failure

“Answer is correct, except with probability 1 in 10,000”

Approximation and Randomization

More generally,

Approximation: find an answer correct within a $(1 \pm \epsilon)$ factor approximation

Randomization: allow a small probability of failure / success probability $(1 - \delta)$

Here we want to develop Approximation **and** Randomization algorithms,

called (ϵ, δ) -approximations.

Approximation

$$\text{True answer} \leq \text{output} \leq \alpha \cdot \text{True answer}$$

Goal: minimize α as much as we can

- Best is $\alpha = 1 + \epsilon$, for all $\epsilon > 0$
- We will require the result to hold with probability of $1 - \delta$.

(if δ is very small then the probability is close to 1).

For example, if $\epsilon = \frac{1}{3}$ and $\delta = 0.01$ then we will have an error of more than $\frac{1}{3}$ only 1% of the time(in 99% we are good!!)